

LASU Certificate Generator

OOP Design Patterns · C++17 · Winsock HTTP Server · Web UI

CSC 226 — Object-Oriented Programming II — Lagos State University — July 2026

Strategy

Template Method

Factory

Inheritance

Polymorphism

RAII

std::map

Multithreading

Group 3: Idowu Fawaz, Idowu Matthew, Ikechukwu Queen, Inedu Esther, Iniotuh Greatest,
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Overview & Architecture

Desktop web app: generate, manage, and print HTML certificates

3 Certificate Types

Excellence, Completion, Participation

3 Templates

Formal (serif)
Modern (sans)
(mono)

CSV Batch Import

File path or paste · 10 sample students

Search, Sort

Name, course, date

Browser



HttpServer

Winsock TCP



CertificateGenerator

Generates HTML

std::map<int, Certificate>

Data flow: Browser → HTTP (JSON) → HttpServer → CertificateGenerator → Certificate

Inheritance & Encapsulation

Person (abstract) → Student (concrete) · All data private

```
class Person {                // Abstract base
protected:
    std::string name_;
public:
    virtual ~Person() = default;
    virtual std::string
        getDescription() const = 0; // pure virtual
};

class Student : public Person { // is-a Person
    std::string course_, grade_;
public:
    double gradeToPoints() const;
    // 'A' -> 4.0  'B+' -> 3.3  '93' -> 3.72
    static bool isValidGrade(const&);
    std::string getDescription()
        const override;
};
```

```
// Encapsulation in Generator
class CertificateGenerator {
    std::map<std::string, Student>
        students_;                // private
    mutable std::mutex mutex_;    // private
public:
    void addStudent(const&);       // validated
    bool removeStudent(const&);
    Student* findStudent(const&);
    void loadFromCsv(const&);
};
```

- **Inheritance:** Student is-a Person — inherits name_, adds course/grade
- **Encapsulation:** All data private — addStudent() validates before storing, throws on bad data

Polymorphism & Abstract Base Classes

3 ABCs, 7 pure virtuals, runtime vtable dispatch

```
class Certificate {           // Abstract base
public:
    virtual std::string
        generateHTML() const;    // Template Method
    virtual std::string
        getTypeName() const = 0; // pure virtual
};

class CertOfExcellence : public Certificate {
    std::string getTypeName() const override
    { return "Excellence"; }
};

class CertOfCompletion : public Certificate {
    std::string getTypeName() const override
    { return "Completion"; }
};
```

- **3 ABCs:** Person, CertificateTemplate, Certificate
- **7 pure virtual methods** total
- Runtime dispatch via vtable
- Client code works with base pointers

```
// Factory returns subclass at runtime
auto tmp1 = makeTemplate(style);
auto cert = makeCertificate(
    type, s, std::move(tmp1));
file << cert->generateHTML();    // polymorphic!
```

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HTTP Server & REST API

Winsock TCP · 12 endpoints · Embedded SPA · Thread safety

```
// Accept loop: each conn on its own thread
while (true) {
    SOCKET client = accept(server,
        nullptr, nullptr);
    if (client == INVALID_SOCKET) break;
    std::thread(handleClient, client,
        std::ref(gen)).detach();
}
```

- All HTML/CSS/JS in C++ raw string literal
- Frontend uses fetch() — no reloads
- Theme toggle: Light/Dark/System

Method	Path	Action
GET	/	Serve Web UI
GET	/list	All students JSON
GET	/stats	Statistics JSON
POST	/add	Insert student
POST	/remove	Delete student
POST	/load	CSV import
POST	/generate	Create certificates
POST	/sort	Sort students

- std::mutex guards map (thread-safe)
- ValidationException → JSON error
- CSV: invalid lines skipped

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